

Advanced (Habanero)

- Q1.** Write the quadratic $f(x) = x^2 + 4x + 4$ in vertex form.
(A) $f(x) = (x + 2)^2$
(B) $f(x) = (x - 2)^2$
(C) $f(x) = x^2 + 4x + 4$
(D) $f(x) = (x + 4)^2$
(E) $f(x) = (x - 4)^2$
- Q2.** The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$. What does h represent?
(A) x -intercept
(B) y -intercept
(C) vertex's x -coordinate
(D) vertex's y -coordinate
(E) axis of symmetry
- Q3.** Which quadratic function has the intercept form $f(x) = a(x - p)(x - q)$, where p and q are the x -intercepts?
(A) $f(x) = x^2 - 7x + 12$
(B) $f(x) = x^2 + 5x - 6$
(C) $f(x) = x^2 - 4x - 3$
(D) $f(x) = x^2 + 2x - 15$
(E) $f(x) = x^2 - 9x + 20$
- Q4.** The quadratic $f(x) = x^2 + 2x - 6$ can be rewritten in vertex form as $f(x) = (x + 1)^2 - 7$. What is the vertex?
(A) $(-1, 7)$
(B) $(-1, -7)$
(C) $(1, -7)$
(D) $(1, 7)$
(E) $(0, -6)$
- Q5.** Write the quadratic $f(x) = x^2 - 5x - 6$ in intercept form.
(A) $f(x) = (x - 6)(x + 1)$
(B) $f(x) = (x + 6)(x - 1)$
(C) $f(x) = (x - 2)(x - 3)$
(D) $f(x) = (x + 2)(x + 3)$
(E) $f(x) = (x + 6)(x + 1)$
- Q6.** The quadratic $f(x) = x^2 - 4x - 3$ has x -intercepts at $x = 3$ and $x = -1$. What is the equation of the axis of symmetry?
(A) $x = -1$
(B) $x = 3$
(C) $x = 1$
(D) $x = 0$
(E) $x = 2$
- Q7.** The quadratic $f(x) = (x - 2)^2 - 9$ can be rewritten in standard form as $f(x) =$
(A) $x^2 - 4x - 5$
(B) $x^2 - 4x + 5$
(C) $x^2 - 2x - 5$
(D) $x^2 + 4x - 5$
(E) $x^2 - 4x + 1$
- Q8.** Which of the following is the vertex form of $f(x) = x^2 + 10x + 16$?
(A) $f(x) = (x + 5)^2 - 9$
(B) $f(x) = (x + 8)^2 - 16$
(C) $f(x) = (x + 6)^2 - 20$
(D) $f(x) = (x + 4)^2 - 0$
(E) $f(x) = (x + 10)^2 - 84$

Open Ended Questions (FRQ)

- Q9.** Write the quadratic function $f(x) = 2x^2 - 12x - 14$ in vertex form and find its vertex.
- Q10.** Find the equation of the quadratic function with x -intercepts $x = 2$ and $x = 5$, and with a y -intercept of 10. Write the equation in both intercept form and vertex form.

Chef's Tips

- Remind students to show all work and explain their reasoning for free-response questions.
- For multiple-choice questions, ensure students select the best answer from the options provided.
- Make sure students are familiar with the different forms of quadratic functions (standard, vertex, intercept) before attempting the test.